Project 01

Secure Enterprise Network — Design, Configuration & Verification

Platform: Cisco Packet Tracer **Status:** Implemented and verified in simulation

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Devices: Cisco 2911 router, two Cisco 2960 switches, four PCs, one server

1. Overview

This project simulates a small enterprise LAN divided into separate VLANs for ADMIN, IT, STAFF, GUEST, SERVERS, and MANAGEMENT. A Cisco 2911 router provides inter-VLAN routing using router-on-a-stick subinterfaces. The router also provides DHCP for user VLANs and applies an extended ACL to block guest traffic to selected internal networks.

The switches use SSH version 2 for remote management, with a standard ACL that permits management access from the IT subnet. Unused switch access ports are administratively disabled. An internal server provides an HTTP intranet page and DNS resolution for `intranet.company.local`.

This is a **Packet Tracer simulation**, not a production deployment or a test on physical hardware.

2. Objectives

- Create departmental VLANs and an IP addressing plan.
- Configure 802.1Q trunks and router-on-a-stick routing.
- Provide DHCP to user endpoints.
- Restrict guest access to selected internal networks.
- Restrict switch SSH management to the IT subnet.
- Disable unused switch ports.
- Configure and verify internal HTTP and DNS services.
- Record repeatable tests and evidence.

3. Topology and cabling

From	To	Purpose
R1 G0/0	SW1 G0/1	Router-to-switch trunk
SW1 G0/2	SW2 G0/1	Inter-switch trunk
SW1 Fa0/2	PC-ADMIN Fa0	ADMIN endpoint
SW1 Fa0/3	PC-IT Fa0	IT endpoint
SW1 Fa0/4	SRV1 Fa0	Server endpoint
SW2 Fa0/2	PC-GUEST Fa0	GUEST endpoint
SW2 Fa0/3	PC-STAFF Fa0	STAFF endpoint

Evidence: `evidence/01-topology.png`

4. VLAN and IP plan

VLAN	Name	Subnet	Gateway
10	ADMIN	`192.168.10.0/24`	`192.168.10.1`
20	IT	`192.168.20.0/24`	`192.168.20.1`

VLAN	Name	Subnet	Gateway
30	STAFF	`192.168.30.0/24`	`192.168.30.1`
40	GUEST	`192.168.40.0/24`	`192.168.40.1`
50	SERVERS	`192.168.50.0/24`	`192.168.50.1`
99	MANAGEMENT	`192.168.99.0/24`	`192.168.99.1`
999	NATIVE_UNUSED	No IP	None

Key addresses observed

Device	Address	Method
R1 VLAN gateways	`192.168.<VLAN>.1/24`	Static
SW1 management SVI	`192.168.99.2/24`	Static
SW2 management SVI	`192.168.99.3/24`	Static
PC-ADMIN	`192.168.10.21/24`	DHCP
PC-IT	`192.168.20.21/24`	DHCP
PC-STAFF	`192.168.30.21/24`	DHCP
PC-GUEST	`192.168.40.21/24`	DHCP
SRV1	`192.168.50.10/24`	Static

The first 20 addresses in each user subnet were excluded from DHCP allocation. DHCP pools serve VLANs 10, 20, 30, and 40. SRV1 uses a static address.

5. Implemented controls and services

VLANs and trunks

VLANs 10, 20, 30, 40, 50, 99, and 999 were created on both switches. Required access ports were assigned to the correct VLANs. Trunks use native VLAN 999 and allow `10,20,30,40,50,99,999`.

Inter-VLAN routing

R1 uses subinterfaces on `G0/0` for VLANs 10, 20, 30, 40, 50, and 99. VLAN 999 is configured as the native VLAN without an IP address.

DHCP

R1 provides DHCP for ADMIN, IT, STAFF, and GUEST. Each pool supplies a gateway and DNS server `192.168.50.10`.

Guest ACL

The extended ACL `GUEST-INBOUND` is applied inbound on `G0/0.40`. It denies guest-sourced traffic to:

- `192.168.10.0/24` (ADMIN)
- `192.168.20.0/24` (IT)
- `192.168.30.0/24` (STAFF)
- `192.168.50.0/24` (SERVERS)
- `192.168.99.0/24` (MANAGEMENT)

A final permit rule allows guest traffic to other destinations not matched above. This does not prove Internet connectivity; no Internet/WAN was configured.

Switch management

SSH version 2 is enabled on SW1 and SW2. VTY lines use local authentication and SSH transport. The standard ACL `IT-MANAGEMENT` permits source addresses in `192.168.20.0/24` and denies other sources. PC-IT successfully SSH-connected to both switches.

Unused ports

Unused access ports were administratively disabled. The required endpoint and trunk interfaces remain connected.

Internal HTTP and DNS

SRV1 is `192.168.50.10/24`, gateway `192.168.50.1`, and DNS server `192.168.50.10`. HTTP and DNS are enabled. The A record `intranet.company.local` resolves to `192.168.50.10`.

6. Verification commands

Run these in the **CLI tab of the relevant Packet Tracer device**. They are Cisco IOS commands, not Ubuntu shell commands.

On R1

```
enable
show ip interface brief
show ip route
show ip dhcp binding
show access-lists GUEST-INBOUND
show running-config
```

On SW1 and SW2

```
enable
show vlan brief
show interfaces trunk
show interfaces status
show ip interface brief
show ip ssh
show access-lists IT-MANAGEMENT
show running-config
```

From PC-ADMIN (Desktop > Command Prompt)

```
ipconfig
ping 192.168.10.1
ping 192.168.20.21
ping 192.168.50.10
```

Then open PC-ADMIN > Desktop > Web Browser and test:

```
http://192.168.50.10
http://intranet.company.local
```

From PC-GUEST (Desktop > Command Prompt)

```
ping 192.168.40.1
ping 192.168.10.21
ping 192.168.50.10
```

Expected: the guest gateway responds; the ADMIN PC and server tests are blocked by the guest ACL.

7. Test results

Test	Expected	Observed	Result
ADMIN DHCP	Address in VLAN 10	`192.168.10.21/24`, gateway `192.168.10.1`	PASS
ADMIN to gateway	Success	4/4 replies	PASS
ADMIN to IT PC	Success	4/4 replies	PASS
ADMIN to server	Success	3/4 replies; first packet timed out	PASS with initial packet loss
HTTP by IP	Page loads	Successful	PASS
HTTP by DNS name	Page loads	Successful	PASS
GUEST to gateway	Success	4/4 replies	PASS
GUEST to ADMIN PC	Blocked	`Destination host unreachable` from `192.168.40.1`	PASS

Test	Expected	Observed	Result
GUEST to server	Blocked	`Destination host unreachable` from `192.168.40.1`	PASS
Switch SSH status	SSH v2 enabled	Confirmed on both switches	PASS
Management ACL	IT subnet permitted; others denied	ACL shown and IT match counters observed	PASS
Required switch links	Connected	Access links and trunks connected	PASS
Unused access ports	Disabled	Confirmed in interface status	PASS

The first ping to the server timed out, while the following three succeeded. This is consistent with initial ARP resolution in a simulator and was not persistent packet loss in this test.

8. Evidence screenshots

Capture the following screenshots from Packet Tracer. Keep screenshots readable and crop out unrelated desktop content.

1. `01-topology.png` — entire topology and device labels.
2. `02-vlan-and-trunks.png` — `show vlan brief` and `show interfaces trunk`.
3. `03-routing-and-dhcp.png` — R1 `show ip route`, `show ip interface brief`, and `show ip dhcp binding`.
4. `04-guest-acl-blocking.png` — guest gateway success and blocked pings to ADMIN/server.
5. `05-ssh-management-acl.png` — `show ip ssh` and `show access-lists IT-MANAGEMENT` on switches.
6. `06-intranet-http-dns.png` — browser showing intranet by DNS name; optionally also capture IP URL.
7. `07-disabled-ports.png` — `show interfaces status` showing unused ports disabled.

Do not include passwords, private keys, or other secrets in screenshots. If a configuration view contains credentials, crop or redact that part.

9. Limitations and possible next steps

- The network is simulated in Packet Tracer; it is not a physical or production deployment.
- No WAN or Internet connectivity is configured.
- The guest ACL blocks specified internal subnets but permits other destinations. A stricter future policy could explicitly permit required guest services and deny other traffic.
- No dedicated firewall, IDS/IPS, centralized logging, high availability, or redundant routing is implemented.
- Tests cover the scenarios listed above, not every possible traffic flow or a full security assessment.
- Further versions could add monitoring, redundancy, more explicit guest policies, and automated configuration checks.

10. Repository structure

```
01-secure-enterprise-network/
■■■ README.md
■■■ 01-Secure-Enterprise-Network.pkt
■■■ configs/
■   ■■■ R1.txt
■   ■■■ SW1.txt
■   ■■■ SW2.txt
■■■ evidence/
■   ■■■ 01-topology.png
■   ■■■ 02-vlan-and-trunks.png
■   ■■■ 03-routing-and-dhcp.png
■   ■■■ 04-guest-acl-blocking.png
■   ■■■ 05-ssh-management-acl.png
■   ■■■ 06-intranet-http-dns.png
■   ■■■ 07-disabled-ports.png
■■■ docs/
    ■■■ test-results.md
```

The `.pkt`, README, and screenshots are the essential first version. Configuration exports are optional; review and sanitize them before publishing.

11. Why this project is useful

This project demonstrates foundational enterprise networking tasks: segmentation, routing, address allocation, access control, secure device administration, service configuration, and verification. It gives a reviewer evidence that the design was tested, not merely drawn.

It is a suitable foundation for future projects involving enterprise WAN routing, branch connectivity, network hardening, monitoring, and automation.

12. Conclusion

The simulated enterprise network successfully separates user groups into VLANs, routes between VLANs, assigns user addresses through DHCP, restricts guest access to selected internal networks, limits switch SSH management to the IT subnet, disables unused ports, and serves an internal webpage through HTTP and DNS.

Artifact: `01-Secure-Enterprise-Network.pkt`

Environment: Cisco Packet Tracer

Status: Selected connectivity, service, and security-policy tests completed.